

DATA ANALYTICS

CUSTOMER INSIGHTS

Customer Shopping Behavior Analysis

Analyzing 3,900 purchases across product categories to uncover spending patterns, customer segments, product preferences, and subscription behavior — guiding strategic business decisions.



The Dataset

3,900 Transactions, 18 Columns

Customer Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Shopping Behavior

Discount, Promo Code, Previous Purchases, Frequency,
Review Rating, Shipping Type

Missing Data

37 values in the Review Rating column — imputed using median rating per product category.



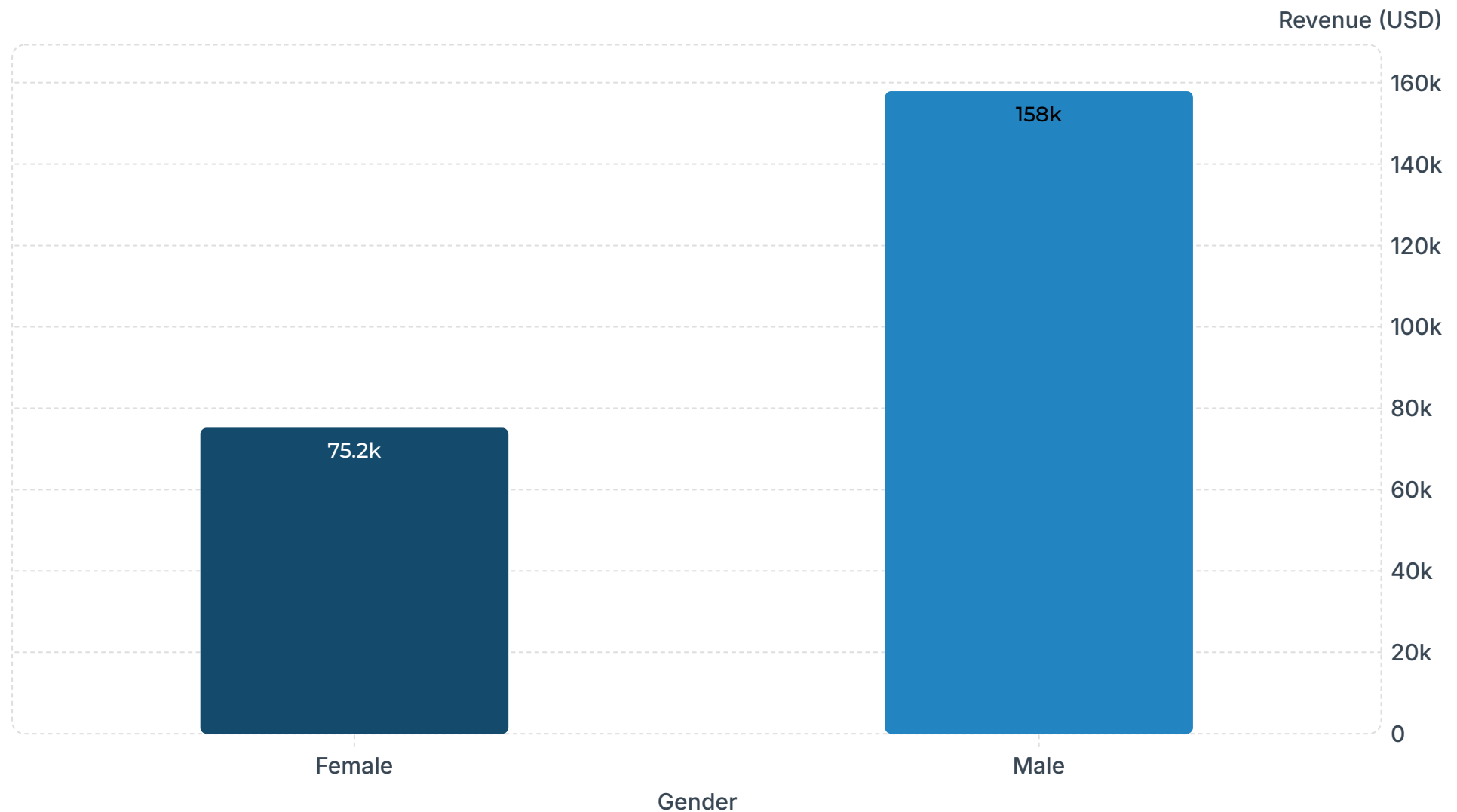
Exploratory Data Analysis

Cleaning & Preparing the Data in Python

Columns standardized to snake case. Redundant promo_code_used column dropped after confirming it duplicated discount_applied.



Revenue & Spending Insights



839

High-Spending Discount Users

Customers who used discounts but still spent above the average purchase amount.

\$60.48

Express Shipping Avg.

vs. \$58.46 for Standard — express users spend slightly more per order.

Product Performance

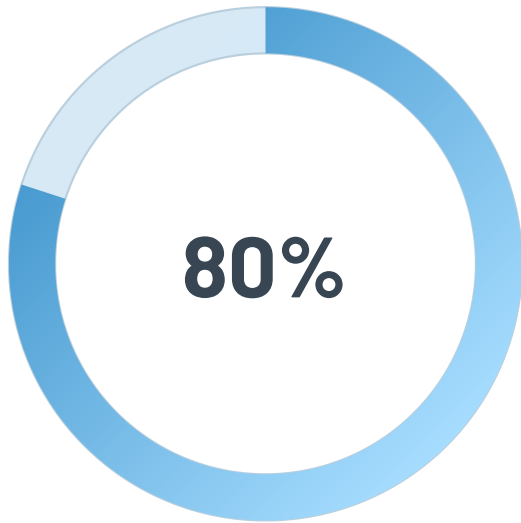
Top-Rated & Discount-Dependent Products

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

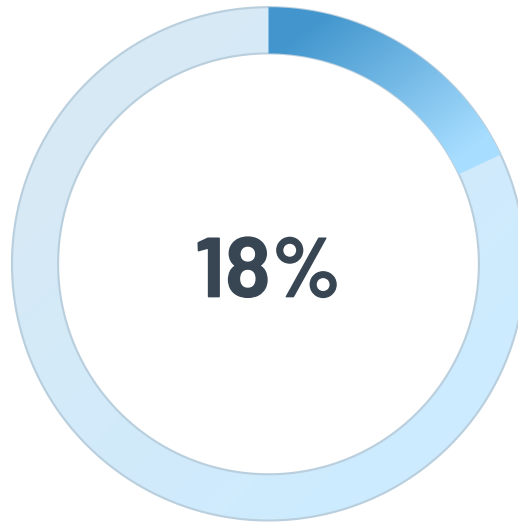
Customer Segmentation

Who Are Our Customers?



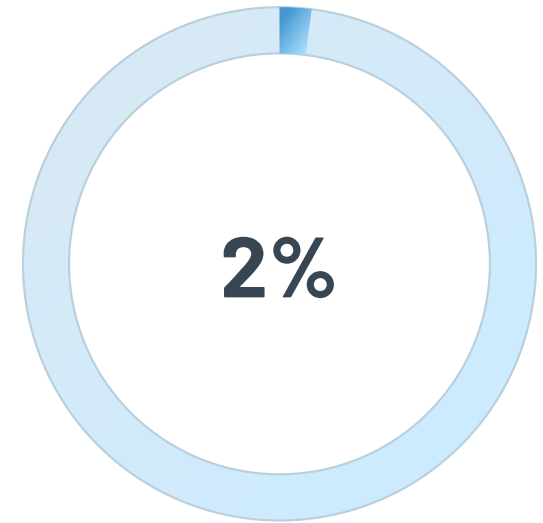
Loyal Customers

3,116 customers with extensive purchase history



Returning

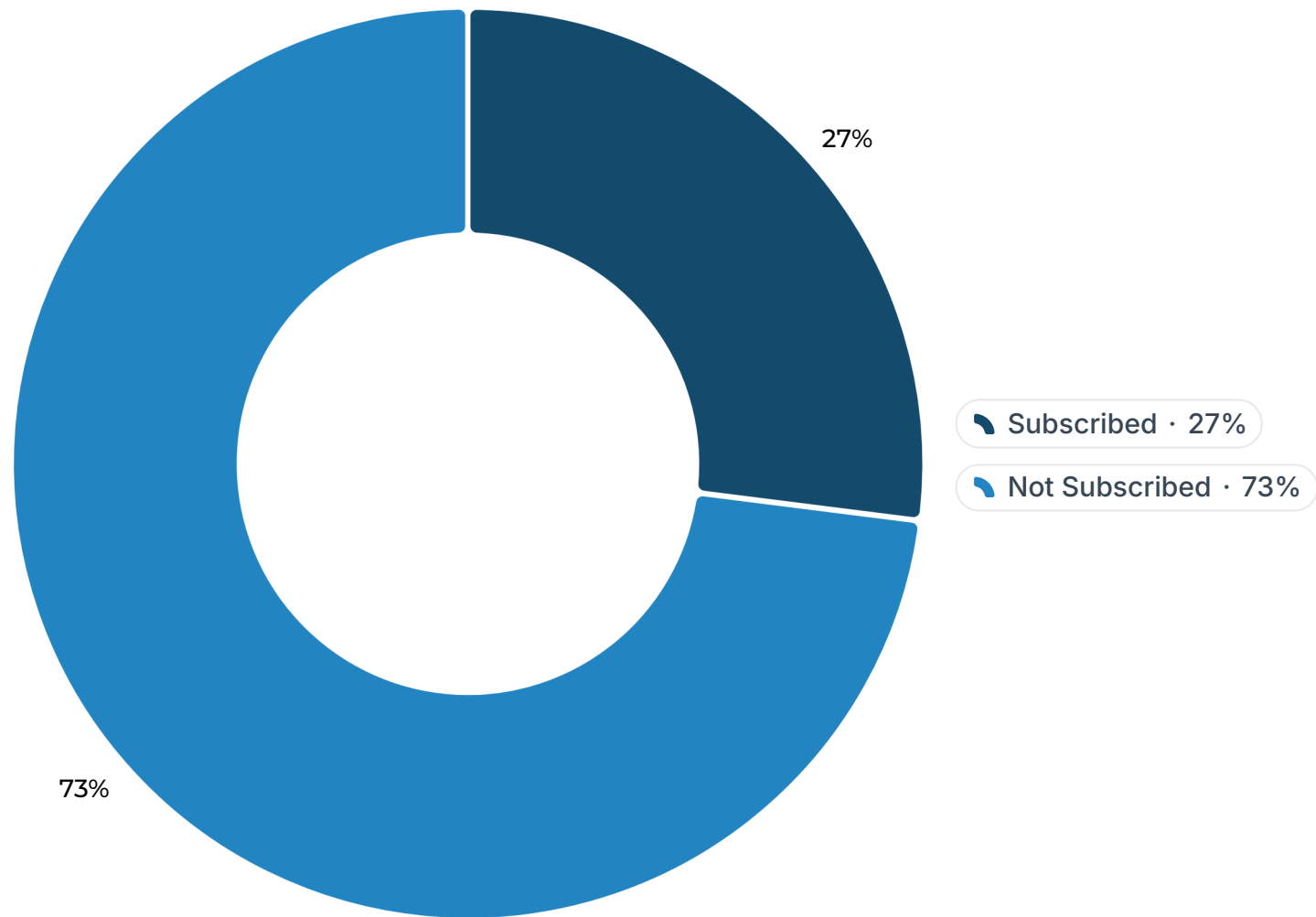
701 customers making repeat purchases



New

83 first-time customers

Subscribers vs. Non-Subscribers



\$59.49

Subscriber Avg Spend

Total revenue: \$62,645

\$59.87

Non-Subscriber Avg Spend

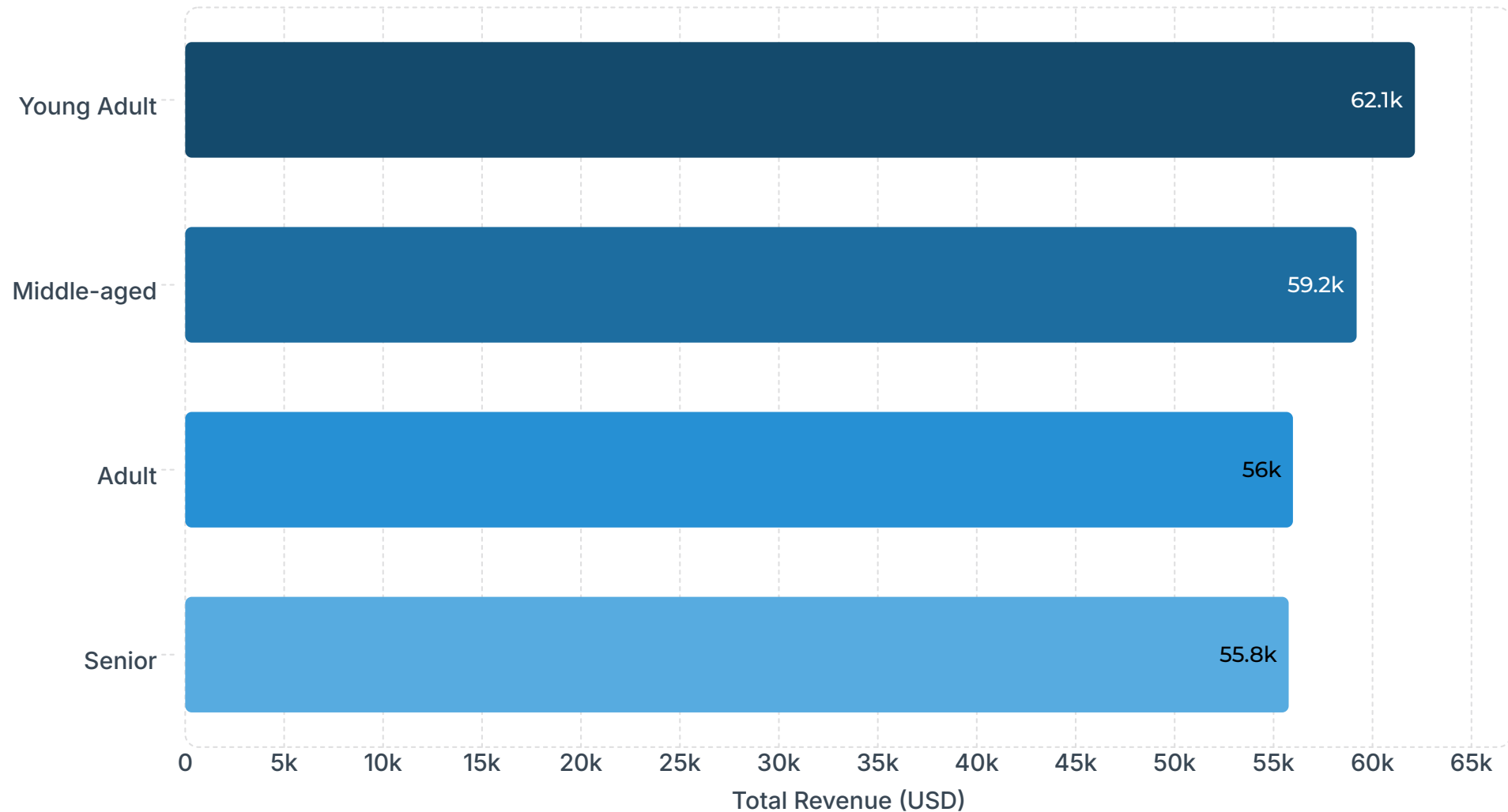
Total revenue: \$170,436

Among repeat buyers with 5+ purchases, 2,518 remain non-subscribers while 958 have subscribed — a significant untapped conversion opportunity.

Revenue by Age Group

Young Adults Lead in Revenue

Age Group



Revenue is relatively evenly distributed across age groups, with Young Adults contributing the most at \$62,143 and Seniors the least at \$55,763.

Power BI Dashboard

Visualizing the Full Picture

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

An interactive dashboard with filters for Subscription Status, Gender, Category, and Shipping Type — surfacing revenue by category, sales by age group, and subscription breakdown at a glance.

Clothing dominates revenue at ~\$100K, followed by Accessories (\$50K), Footwear (\$25K), and Outerwear (\$10K).



Strategic Next Steps

Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers to convert the 2,518 non-subscribed repeat buyers.

Loyalty Programs

Reward repeat buyers to move them into the Loyal segment.

Review Discounts

Balance sales boosts with margin control on discount-dependent products.

Product Positioning

Highlight top-rated and best-selling products in campaigns.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.